Classification of the Arctic Ice Types (AIT) Using the Nadir-Observing Radiometer and Altimeter Data From the Chinese Marine Dynamic Environment Satellite HY-2B

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***Abstract*—The passive calibration microwave radiometer (CMR) and active radar altimeter (RA) microwave data from the Chinese marine environmental dynamics satellite HaiYang-2B (HY-2B) were utilized, for the first time, to classify Arctic first-year ice (FYI) and multiyear ice (MYI) throughout the entire calendar year of 2019. We have applied two variants of the K-means algorithm, called K-means-nearest (KMN) and K-means-argmax (KMA), to classify Arctic ice types (AIT). KMA proved to be more stable. By combining both active and passive sensor data from the satellite's nadir points and employing the KMA algorithm, the overall accuracy of ice-type identification exceeds 80% when compared to reference data from OSI-SAF. This method provides a significant improvement compared to using only single-sensor data. The value of the K-means parameter K was varied to find optimal values for stable classification. Through the analysis of total precipitation, snowfall, and clear-sky direct solar radiation at the sea/ice surface level from the ERA5 atmospheric reanalysis model dataset, it was observed that the total precipitation and snowfall in the polar region have less impact on identifying ice types during winter and more during the early stages of sea ice growth.**

***Index Terms*—Arctic ocean, classification, K-means, microwave, radar altimeter (RA), radiometer, sea ice types.**

I. Introduction

SEA ice is an essential component of the Earth's climate system, which is crucial in justifying global temperatures and ocean circulation [1], [2], [3]. Different types of sea ice have unique physical, chemical, and biological properties that influence their growth, melting, deformation, and motion [4], [5]. Understanding the variety of sea ice types is critical for predicting the effects of climate change and managing Arctic ecosystems.

Currently, many remote sensing instruments are available for sea ice-type identification, which can be seen as an inversion process. Such instruments are, e.g., scatterometers [6], [7], [8], [9], SAR [10], [11], [12], and radar altimeters (RAs) [13], [14]. Advantages of different space-borne instruments are related to their different detection capabilities. With the rapid development of remote sensing technology for observing the cryosphere, several representative sea ice-type products based on remote sensing data have gradually emerged in the past few decades.

Microwave instruments currently play an important role in the inversion of sea ice parameters. They can penetrate the cloud cover, operate without solar illumination, and offer a wide field of view. These instruments can be used as standalone instruments or jointly. Such instruments usually use a large observation-angle range to cover a wide across-track area [15], [16], [17]. The joint utilization of small-observation-angle space-borne remote sensing instruments (e.g., RA and radiometer designed for atmospheric correction) has been applied to sea ice cover in the Caspian and Aral Seas as well as to ice and snow over large Boreal lakes [18], [19], [20]. A comprehensive study using this meteorology for polar sea ice retrieval, particularly in extracting information on sea ice type and age, is still limited. This is primarily due to the inherent challenges associated with sea ice environments, including the high spatial and temporal variability of ice conditions, complex surface scattering mechanisms of radar signals, and the difficulty of identifying ice types using only a single microwave instrument.

Although the instrument combination proposed in this study has a narrower swath width than many other satellite sensors, it offers high resolution along the swath track and unique simultaneous active–passive observing capabilities. These characteristics enable more comprehensive sea ice monitoring.

The Chinese marine environmental dynamics satellite Haiyang-2B (HY-2B) has the capability to provide a combination of RA and calibration microwave radiometer (CMR) missions [21], [22]. HY-2B offers a unique opportunity by simultaneously acquiring RA and nadir-looking microwave

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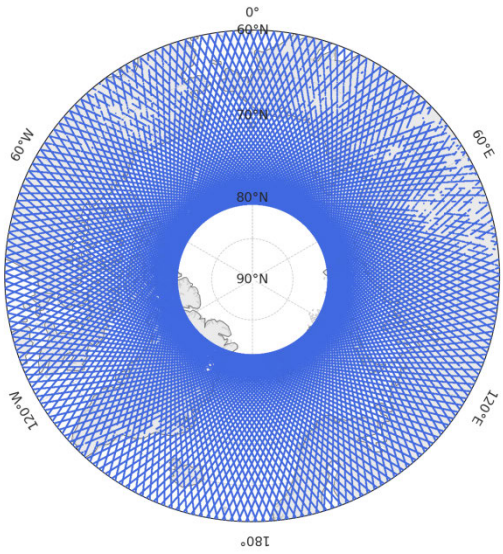


Fig. 1. Observation trajectory of the HY-2B Satellite subsatellite during a repeat cycle of 14 days.

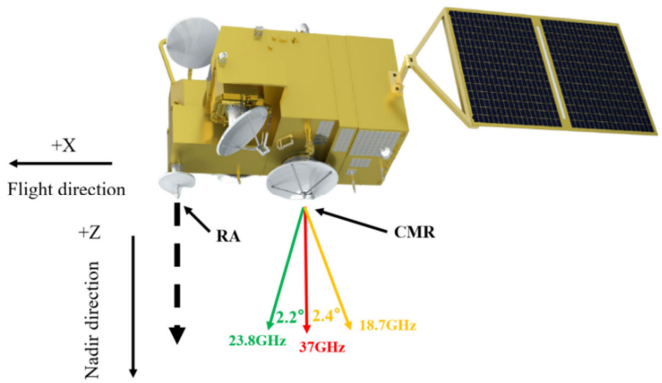


Fig. 2. Schematic of the pointing antenna beam directions of the CMR and RA.

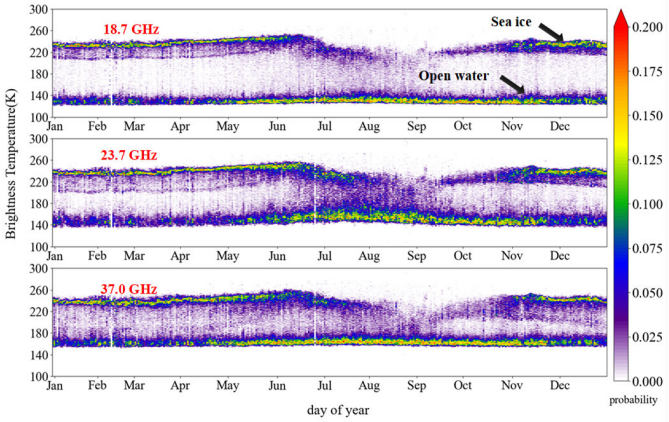


Fig. 3. Probability distribution time series illustrating the distribution of the brightness temperature.

radiometer data over oceanic and polar regions. This integrated dataset can potentially improve the discrimination of different sea ice types by leveraging both active and passive sensing characteristics.

This study aims to investigate the potential use of machine learning methods for classifying sea ice types using the combined RA and CMR data observed by HY-2B. We have developed two different methods based on the K-means clustering algorithm for sea ice-type classification. To the best of our knowledge, this is the first time such a combination of HY-2B data has been used for sea ice classification. We are not aware of any similar previous work that applied data from small-observation-angle space-borne remote sensing instruments for Polar sea ice applications.

The objectives of this study are: 1) to explore the nadir remote sensing observations for sea ice applications; 2) to identify and optimize a simple machine learning method; and 3) to classify the first-year ice (FYI) and multiyear ice (MYI) over an annual cycle. We expect to provide an alternative remote sensing product using HY-2B data to discriminate Arctic FYI and MYI. Based on the results of this study, this new type of sea ice remote sensing product provides additional value.

II. Data and Data Characteristics

A. RA and CMR

The HY-2B satellite platform is equipped with an array of instruments designed for ocean observation, including a RA, a microwave scatterometer, a scanning microwave radiometer, and a three-frequency CMR. The primary focus of this study involves the RA and CMR. Operating on a solar-synchronous, frozen orbit, the HY-2B satellite platform has a repeat cycle of 14 days. Fig. 1 illustrates the flight tracks of the HY-2B platform in the northern polar region over one 14-day period. Due to the observation inclination angle of the satellite platform being 99.34015°, the satellite trajectory is able to cover the majority of the polar regions.

CMR on the HY-2B satellite acts as a nadir-looking three-frequency microwave radiometer. In Fig. 2, the observational geometries of the CMR and RA are depicted. The CMR operates at the frequencies of 18.7, 23.8, and 37.0 GHz, each with the observation angles of 2.2°, 2.4°, and 0°, respectively (see Fig. 2 for details; source: http://www.nsoas.org.cn/). The RA, similar to CMR, is also a nadir-looking instrument designed for Earth observation, with both Ku (15 GHz) and C (5 GHz) bands available.

In this study, data from regions north of 60° of northern latitude were utilized.

The HY-2B RA is a nadir-looking dual-frequency RA operating at Ku band and C band. Its footprint is approximately circular, with a diameter of about 1.9 km. In this study, the RA Level-1B products were used to extract the backscatter coefficient σ^0 for both frequency bands. The Ku- and C-band σ^0 values were obtained from the calibrated echo information provided in the Level-1B products and are hereafter referred to as Ku_σ^0 and C_σ^0, respectively.

Before classification, the RA observations were quality-controlled. Records with missing geolocation information, missing Ku- or C-band σ^0 values, abnormal quality flags,

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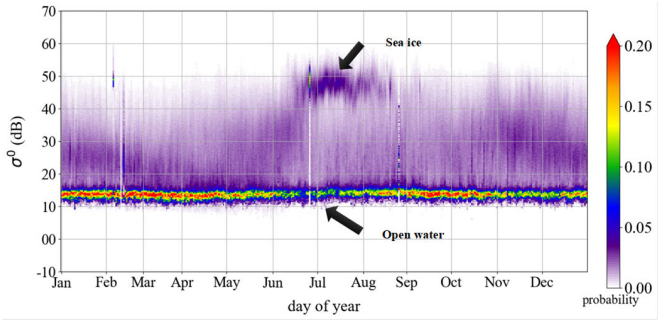


Fig. 4. Probability distribution time series plot for ku band σ^0.

or physically unreasonable σ^0 values were removed. Observations outside the study region north of 60°N were also excluded. The remaining valid σ^0 measurements were then collocated with the HY-2B CMR brightness-temperature observations to construct the active–passive feature vectors used for sea ice-type classification.

B. Time Series of Brightness Temperature and σ^0

Daily probability distributions were estimated for the CMR brightness-temperature values from the three bands and for the RA backscatter coefficient (σ^0). By compiling the daily results from an entire year, we obtained the annual probability distribution time series shown in Figs. 3 and 4.

In general, as shown in Fig. 3, distinguishing between ice and water is obvious for all three brightness-temperature bands, emphasizing the capability of brightness temperature to discriminate between ice and water consistently throughout the year. Distinct stratification is observed within clusters seen in the sea ice distribution, with the most prominent occurring from January to May at 23.8 GHz. These stratified areas within the sea ice clusters indicate different types of sea ice. However, during the melting season (between June and October), the sea ice clusters gradually become more scattered and get mixed with seawater clusters. Simultaneously, the various types of sea ice clusters undergo a gradual mixing, making separation between the ice types and water more challenging.

Similar to the patterns observed in the brightness-temperature bands, the separation between sea ice and open water (OW) σ^0 clusters remains throughout the year, albeit with less prominent clustering of sea ice. The notable difference in the distribution compared to the brightness-temperature band distributions is the good separation of sea ice clusters from water clusters during the summer season. This phenomenon establishes a foundation for the collaborative utilization of brightness temperature and backscatter coefficient values for efficient distinguishing between sea ice and OW during the sea ice melt season.

C. Sea Ice Type

To assess the classification performance, the classified HY-2B data are compared against reference data from the Ocean and Sea Ice Satellite Application Facility (OSI-SAF), coordinated by the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT), providing sea ice-type data at a resolution of 10 km [23]. The sea ice types identified include first-year ice (FYI) and multiyear ice (MYI), distinguished based on variations in ice surface roughness and brightness temperature. The OSI-SAF sea ice-type product (OSI-403-d) utilizes a combination of passive and active microwave satellite observations. Passive microwave data are acquired from AMSR2 onboard GCOM-W1 and SSMIS onboard DMSP-F18. These sensors provide brightness-temperature measurements, from which parameters such as the Gradient Ratio (GR1937: the ratio of brightness temperature at 19 GHz to that at 37 GHz) are derived to distinguish emissivity differences between FYI and MYI. Active microwave data are provided by the ASCAT scatterometers onboard the Metop-B and Metop-C satellites, which measure backscatter coefficients sensitive to surface roughness variations. A Bayesian multisensor approach integrates these passive and active inputs to estimate the probability of each sea ice type. This methodology enables consistent and reliable classification of sea ice types on a global 10 km daily grid. In addition, the dataset identifies areas of OW. OSI-SAF data serves as essential reference information in the estimation of sea ice types.

The OSI-SAF product also includes an "ambiguous ice" category, representing sea ice pixels for which a reliable distinction between FYI and MYI cannot be made. In this study, these pixels were excluded from the training and validation datasets during the freezing periods to avoid introducing uncertain FYI/MYI reference labels. During the melting period, when only binary ice/water classification was performed, ambiguous ice pixels were included in the ICE class together with FYI and MYI.

D. ERA 5 Model Data

In this study, we incorporate atmospheric parameter data from the ERA5 climate reanalysis model [24] to examine the factors influencing our classification accuracy. ERA5, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), represents the latest generation of global modeled reanalysis datasets. It encompasses numerous commonly used surface and atmospheric variables spanning from 1950 to the present. Through assimilating extensive ground-based observations, atmospheric-sounding data, and remote sensing data, ERA5 offers a wealth of essential parameters facilitating the exploration of climate change(online: https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5)

III. Methods

In this study, we chose K-means as the basic algorithm for the classification. Meanwhile, on this basis, two new methods were proposed for the research of sea ice classification. The K-means algorithm has found applications across various scientific fields [25]. Despite being originally proposed over 60 years ago [26], K-means has remained a widely employed algorithm in unsupervised classification processes. The fundamental concept of the algorithm involves a random selection of K initial clustering centers from the dataset. Subsequently, the algorithm calculates the distances (typically Euclidean

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TABLE I
CALCULATION PARAMETERS FOR CLASSIFICATION

Name	Unit
Backscatter coefficient of RA (σ^0) Ku band	dB
Backscatter coefficient of RA (σ^0) C band	dB
Brightness temperature of CMR (18.7 GHz)	K
Brightness temperature of CMR (23.8 GHz)	K
Brightness temperature of CMR (37.0 GHz)	K
Gradient Ratio (18.7/37.0 GHz)	—

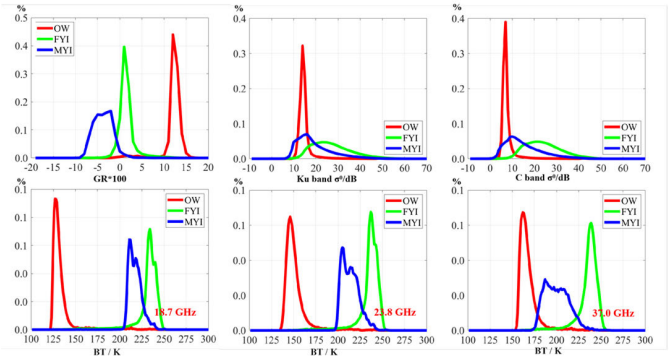


Fig. 5. Probability distributions for OW, FYI, and MYI are illustrated for the six parameters of Table I. The distributions derived from the RA and CMR for OW are depicted in red, FYI in green, and MYI in blue. BT refers to brightness temperature.

distance) between each element and the centers, then assigns them to the nearest cluster. The average values of each of the K categories are then computed to establish new cluster centers, and this iteration continues until the centers no longer undergo changes [27].

A. Parameters Selection

We chose to utilize the brightness temperature values from the CMR, the backscatter coefficient (σ^0) value from the RA, and the gradient ratios (GRs), which have proved to perform well in the classification of sea ice types [28]. The formula for the GR is as follows:

$$\mathrm{GR}(\mathrm{f1},\mathrm{f2}) = \frac{(BT(f2) - BT(f1))}{(BT(f2) + BT(f1))}. \quad (1)$$

The $BT(f1)$ represents the brightness-temperature value at the 18.7-GHz band, and the $BT(f2)$ represents the brightness-temperature value at the 37.0-GHz band. The parameters or features studied for the classification are listed in Table I.

To verify the ability of each parameter to distinguish the MYI, FYI, and OW, we generated single-feature probability distribution plots using data covering the entire year of 2019. The labels for different ice types were defined using OSI-SAF ice-type data.

The probability distributions of the six parameters for the three distinct categories (OW, FYI, and MYI) are presented in Fig. 5. In the probability distributions, greater separation between distributions of different surface types indicates easier classification of the surface type. For all parameters, the OW probability density functions (PDFs) are notably distinguished from the other two classes. Except for the σ^0 values of the two bands, clear thresholds to distinguish OW from ice (FYI and MYI) can be identified in the distribution plots.

However, seasonal variations inevitably induce changes in the features, necessitating a division of the year into reasonable periods for separate classification parametrization.

The seasonal division was based on the daily mean near-surface air temperature averaged over the region north of 80°N. This region was selected to reduce the influence of land areas and islands surrounding the Arctic Ocean and to better represent the large-scale thermal condition of the central Arctic sea ice cover. The temporal variation in daily mean air temperature north of 80°N is shown in Fig. 6, which provides the basis for dividing the annual cycle into different seasonal periods. The 0 °C threshold was used as a practical indicator of the freeze–melt transition. When air temperature approaches or exceeds 0 °C, surface melting, wet snow, and melt pond formation can substantially change the microwave emissivity and radar scattering properties of sea ice. These changes directly affect both CMR brightness temperature and RA σ^0.

It should be noted that air temperature is not the only factor controlling the seasonal evolution of microwave signatures. Surface albedo, melt onset, snow wetness, surface temperature, precipitation, snowfall, and radiation-related variables may also influence the classification results. Therefore, the temperature-based division used here should be regarded as a practical seasonal stratification strategy rather than a unique optimal criterion. The results are expected to be more robust during stable freezing periods, while larger uncertainty may occur near the melting and refreezing transitions.

Typically, a decrease in temperature is a significant factor influencing the transition from ice to water. In this article, we adopt the widely used threshold of 0 °C, representing the freezing temperature of pure water, to divide the one-year period into different subperiods based on sea ice stage of development, specifically analyzing the effects during periods below 0 °C. The division became as follows: days 1–148 (referred to as Period 1), 149–259 (Period 2), and 260–365 (Period 3). This division is generally consistent with the time when the OSI-SAF ice-type data failed the ice-type identification algorithm due to climate warming. Due to the unavailability of ice-type classification results from OSI-SAF for days 149–258, our focus was solely on sea ice-OW classification during this period.

Overall, the spread of the σ^0 values for the two bands concerning first-year ice (FYI) and multiyear ice (MYI) is extensive. When comparing Period 1 (until day 148) with Period 2 (from day254), all three brightness-temperature bands, as well as the GR values and the backward scattering coefficient values, demonstrate the capacity to distinguish between the two sea ice types. The overlap between the two

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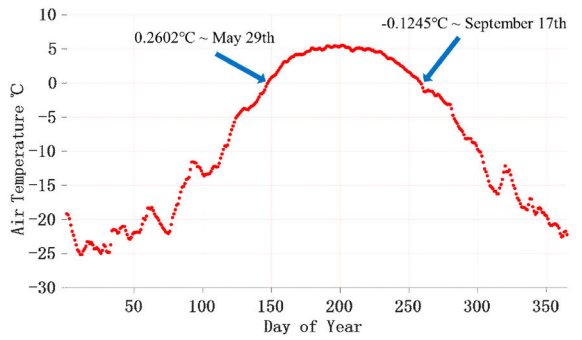


Fig. 6. Daily average air temperature north of the latitude 80 N.

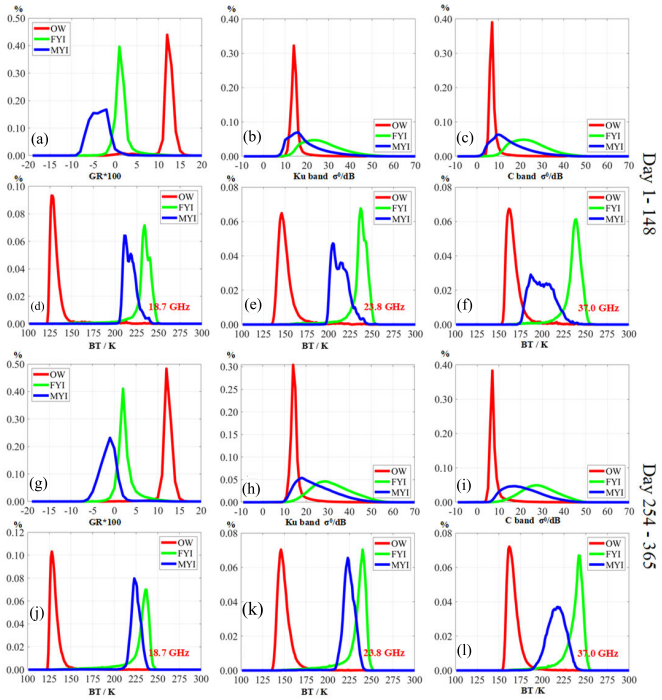


Fig. 7. Probability distribution for OW, FYI, and MYI for the six parameters for the days 1–148 days (a–f) and for the days 259–365 (g–l).

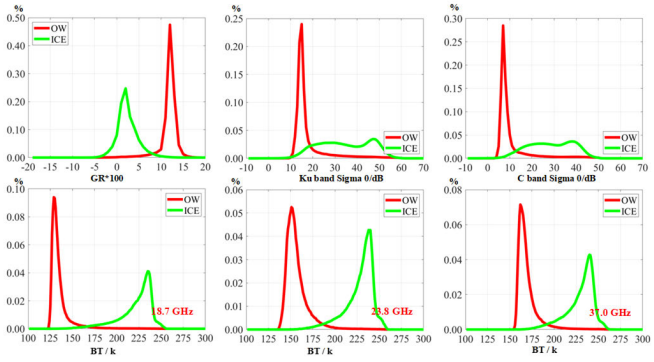


Fig. 8. Probability distributions for OW and ICE (FYI + MYI) for the six parameters and days 149–258.

sea ice types is smaller during Period 1 compared to Period 2; the most pronounced distinction is observed in the brightness-temperature band.

The distributions of the selected parameters for open water (OW), first-year ice (FYI), and multiyear ice (MYI) during the two freezing periods are shown in Fig. 7, while Fig. 8 presents the corresponding parameter distributions for open water and sea ice during the melting period. This enhanced differentiation between ice and water proves to be beneficial for identifying ice and water regions during sea ice melt.

The classification is conducted using data collected above 60°N. The OSI-SAF sea ice-type product used in this study is provided on a 10-km polar stereographic grid and serves as the reference dataset for FYI and MYI classification. The correct classification is determined by the number of measurements classified for each ice type by HY2B, aligning with the ice-type classification from OSI-SAF. HY-2B observations, which have a footprint comparable to but slightly coarser than the OSI-SAF grid spacing, are spatially matched to the OSI-SAF ice-type grid using a nearest-neighbor approach. FYI and MYI masks are then applied to the classified data, and the percentage of FYI and MYI observations with ambiguous and unclassified instances based on thresholds is calculated for each area. The percentage of accurately classified echoes is presented in a manner similar to [13].

The probability distributions of OW and ice types (FYI and MYI) exhibit a distinct log-normal pattern across different time periods. Furthermore, the GR values and σ^0 values for both bands approximately represent a normal distribution. Due to a multitude of external influences in real-world observations, their distributions are not exactly perfect mathematical log-normal or normal distributions. Thus, we apply mathematical fitting methods to approximate the theoretical distributions, used as estimates of class probability distribution functions for these datasets.

B. Correlations Between Various Parameters and Sea Ice Types

To gain a deeper understanding of the correlations between various parameters and sea ice types, we employed the random forest tree classification method [29], known for its high data compatibility, robustness, ease of operation, and capability to efficiently process large datasets. This approach was utilized to assess the correlations between the six parameters and the ice type to be classified during the distinct time periods. This classifier takes the six parameters listed in Table I as its input. The random forest tree classifier was configured with 100 trees, and the tree depth parameter was set to one. Instead of using the random forest trees for direct category classification, we want them to obtain an importance ranking of different parameters for each category. The mean decrease in impurity (MDI) method (e.g., Louppe [30]) is utilized within the context of random forest algorithms to provide the significance of parameters. Its foundational principle posits that the relative ranking (i.e., the depth) of features serving as decision nodes in a decision tree is indicative of the feature's relative importance in forecasting the outcome variable.

Overall, Fig. 9 indicates that the relative importance of the HY-2B features depends strongly on both the seasonal stage and the classification objective. During the freezing periods, the brightness-temperature channels, especially TB18.7 and TB23.8, are the most important features for the OW/FYI/MYI classification. This suggests that passive microwave emission provides stable information on the emissivity contrast among

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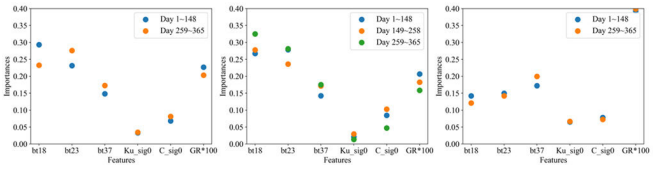


Fig. 9. Importance distribution of parameters. (a) OW, first-year ice, and multiyear ice. (b) OW and sea ice. (c) First-year ice and multiyear ice.

OW, FYI, and MYI under cold and relatively dry surface conditions.

For the OW/sea ice classification, especially during the melting period, the importance of RA backscatter increases. In particular, C-band σ^0 becomes more important, indicating that active microwave backscatter provides complementary information on surface roughness and scattering properties when the brightness-temperature contrast is reduced by surface melting, wet snow, and melt ponds.

For the FYI/MYI classification during the freezing periods, both brightness-temperature and gradient-ratio features contribute to the discrimination of ice types. The GR provides additional spectral information related to the emissivity difference between FYI and MYI, while σ^0 contributes information on surface and near-surface scattering properties.

These results support the feature-selection strategy used in this study. Brightness temperature is the primary feature group for ice-type discrimination during freezing periods, σ^0 provides important complementary information for ice/water separation and surface-condition changes, and GR helps describe spectral emissivity differences. Therefore, the combined use of TB, σ^0, and GR is more appropriate than using a single feature group, and this provides the basis for the multifeature classification framework.

C. Classification Methods

We are working with a sizable dataset, necessitating consideration for computational load and generalization performance. To avoid potential overfitting resulting from an excessively large training dataset, we have chosen to split the data into training and testing datasets with a training dataset size proportionally slightly larger than the testing dataset size. To achieve this, we divide the data into two (4:6) parts while ensuring they possess similar distribution characteristics. Then, 60% of the data are randomly selected to form the training dataset, which is used to train the model. The remaining 40% of the data are assigned to the testing dataset, which is used to evaluate the model's performance. The random division ensures that both datasets are representative of the overall data distribution, reducing potential bias and helping to generalize the model to unseen dataset.

For the ice-type classification using the K-means algorithm, we selected brightness-temperature data from three bands and σ^0 values from two RA bands as parameters. As outlined in Section III-A, the K-means classification categorizes HY-2B data into three classes (OW, FYI, and MYI) during the period of frozen sea ice and two classes (OW and ICE) during the sea ice melting period.

In this section, two K-means-based approaches, namely the K-means-nearest (KMN) and the K-means-argmax (KMA), are used for the identification of sea ice types and OW. In both methods, K-means clustering is first applied to the training dataset, and each cluster is assigned a physical class label according to the class distribution of the reference sea ice-type labels within that cluster. The main difference between KMN and KMA lies in the way a testing sample is assigned to a labeled cluster. In KMN, a testing sample is assigned to the nearest labeled cluster center based on Euclidean distance. In KMA, PDFs are fit for the labeled clusters using the training dataset, and a testing sample is assigned to the labeled cluster with the maximum fit pdf value. The final class of the sample is then determined by the class label of the selected cluster. The PDFs were estimated from the training dataset, and the training and classification procedures are described in detail below.

Training Process:

1) The input features were selected based on the feature-importance analysis described in Section III-B. The selected features include the CMR brightness temperatures, the Ku- and C-band RA backscatter coefficients, and the GR. Samples with missing values or invalid observations were removed before training. Because the selected variables have different units and dynamic ranges, all input features were standardized using the mean and standard deviation calculated from the training dataset. The same normalization parameters were then applied to the testing dataset.
2) The cluster number K was determined by varying K over a predefined range and evaluating the stability of the classification accuracy. Low K values produced large fluctuations in the classification results, whereas K values between 25 and 35 gave more stable outputs. Therefore, K was selected within this stable range for the final classification experiments.
3) K-means clustering was applied to the normalized training dataset. To reduce the influence of random initialization, the clustering was repeated with different initial cluster centers, and the solution with the minimum within-cluster sum of squares was retained. Each training sample was then assigned to one of the K clusters.
4) The physical label of each cluster was determined using the reference sea ice-type labels. The class distribution within each cluster was calculated by mapping the labeled training samples to the corresponding clusters. Each cluster was then assigned to the dominant class by majority voting. During the freezing periods, the cluster labels were assigned as OW, first-year ice (FYI), or multiyear ice (MYI). During the melting period, FYI and MYI were merged into a single sea ice class, and the cluster labels were assigned as OW or sea ice.
5) For the KMA method, parametric PDFs were fit to the observations associated with each labeled cluster. For each selected feature, either a normal or log-normal distribution was used according to the empirical distribution characteristics of the clustered samples. The pdf parameters were estimated from all training samples

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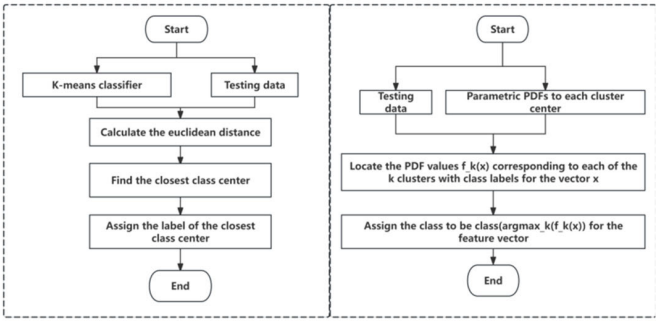


Fig. 10. Flowchart of the KMN (left) and KMA methods (right).

assigned to the corresponding cluster. For a testing feature vector x, the fit PDFs were used to calculate the probability density value fk(x) for each labeled cluster k.

Classification Methods:

KMN:

1) Find the closest class center (according to the Euclidean distance) for a feature vector;
2) Assign the label (class) of the closest class center to the feature vector x.

KMA:

1) Locate the pdf values f_k(x) corresponding to each of the k clusters with class labels for the vector x.
2) Assign the class to be class(Argmax_k(f_k(x)) for the feature vector.

In the classification process presented in this article, seawater and sea ice are initially distinguished, followed by a subsequent classification of the sea ice into the two different ice types. The specific steps for both classification tasks are identical, and the detailed classification flow is provided in Fig. 10.

IV. Results

To evaluate the SI type and OW classification results, we use three evaluation measures: accuracy (A), precision (P), and recall rate (R).

Accuracy is defined as the number of correctly predicted samples in all categories divided by the total number of samples. It is a measure of the overall performance of all categories. n is the number of categories. is the true instance of category. N is the total number of samples

$$\boldsymbol{A} = \frac{\sum_{i=1}^{n} TP_i}{N}. \tag{2}$$

Precision is defined as the number of true positives (TP_i) over the number of true positives plus the number of false positives (FP_i). It measures the reliability of positive classification results. This is computed as an average over all the n categories (classes)

$$\boldsymbol{P} = \frac{1}{n}\sum_{i=1}^{n} \frac{TP_i}{TP_i + FP_i}. \tag{3}$$

Recall is the number of true positives (TP_i) divided by the sum of the sum of and the number of false negatives. It measures the ability to correctly identify the instances of each class; it represents the proportion of actual positive, i.e., correctly classified, instances. Recall is also computed here as an average over all the categories:

The classification performance of the KMN and KMA methods across the three seasonal periods is compared in Fig. 11. Both methods achieved generally good classification results, although their performance varied with the number of clusters (K). In particular, the classification accuracy at the three periods was characterized by oscillations at small K values ($\sim<$ 10), followed by gradual smoothing. The accuracy rates remained above 85%; in terms of precision, the accuracy of the KMA method was lower than that of the KMN method, but still overall higher than 80%. The recall rate was between 80% and 95%, and for all three time periods, there was a significant fluctuation in the K-value range between 5 and 20. Among the three time periods, the recognition accuracy for ice and water classes was slightly higher in summer compared to the freezing period.

The KMN was more volatile to the change of K-value than the KMA method. For Period 1, the KMN method had poorer results for K between 10 and 20, below 60% for multiple values of K. The KMA method significantly outperformed the voting method in multiyear ice identification.

In addition to this, we also classified the data using brightness-temperature values and σ^0 values separately.

The KMN method and KMA schemes are able to obtain good overall accuracy rates using only the brightness-temperature value. However, the classification accuracy of multiyear ice in both schemes varies significantly with the change of K value. In Argmax's scheme, the maximum classification correctness between adjacent K values is about 20% in Period 3. In the KMN method, it varies drastically for Periods 1 and 3 when the K values are small, with the maximum difference in classification correctness between neighboring K-values being about 20%, thus making it difficult to select the most suitable K-value.

The identification of multiyear ice is poor when only σ^0 data are used, as shown in Figs. 5 and 7, the distribution of σ^0 values reveals a weak separation between multiyear and first-year ice. The accuracy results of multiyear ice classification were below 50% for most K values.

For the selection of the K-value, we followed several specific principles: the need for a high rate of classification accuracy, the need for stable classification accuracy, and the need for good evaluation results across multiple evaluation parameters. For these principles of choosing the K-value, the scheme of using the brightness-temperature value in combination with the σ^0 value is clearly more stable and has a better classification accuracy for each classification category than using them alone. The KMA method provides better stability in the classification of multiyear ice compared to the KMN method. In addition, the accuracy rate of multiyear ice classification using the KMA method shows a gradual increase as a function of an increasing K value. This aligns with the general theoretical idea that a more detailed classification leads to a higher accuracy rate.

In the selection of the K value, we chose the value of 30 as an example of K in the range 25–35. Choosing K values

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TABLE II

CLASSIFICATION PERFORMANCE OF THE KMN AND KMA METHODS USING DIFFERENT FEATURE COMBINATIONS DURING THE THREE TIME PERIODS. NOTE: IN PERIOD 2, FYI AND MYI ARE MERGED INTO A SINGLE SEA ICE CLASS

Period		OW	FYI	MYI	Accuracy	Precision	Recall
Day 1~148 (Period 1)	KMN_all	94.6%	97.5%	81.4%	95.4%	91.8%	91.1%
	KMN_bt	95.3%	96.8%	88.1%	95.7%	92.0%	93.4%
	KMN_σ^0	91.4%	97.6%	0.0%	88.4%	58.4%	63.0%
	KMA_all	96.3%	90.3%	92.7%	92.2%	83.1%	93.1%
	KMA_bt	96.5%	90.8%	87.7%	92.2%	83.1%	91.6%
	KMA_σ^0	87.4%	80.3%	30.9%	78.5%	63.3%	66.1%
Day 149~258 (Period 2)	KMN_all	96.2%	95.5%		95.9%	95.7%	95.8%
	KMN_bt	96.3%	95.1%		95.8%	95.6%	95.7%
	KMN_σ^0	86.9%	90.6%		88.4%	88.0%	88.7%
	KMA_all	92.5%	98.4%		95.0%	94.6%	95.5%
	KMA_bt	92.3%	97.3%		94.4%	94.0%	94.8%
	KMA_σ^0	80.2%	94.8%		86.3%	86.5%	87.5%
Day 259~365 (Period 3)	KMN_all	98.6%	93.8%	82.0%	95.0%	91.8%	91.5%
	KMN_bt	98.8%	93.3%	87.8%	95.6%	92.7%	93.3%
	KMN_σ^0	96.2%	92.6%	19.9%	86.4%	78.6%	69.6%
	KMA_all	98.9%	88.3%	82.4%	93.2%	88.1%	89.9%
	KMA_bt	98.8%	83.5%	81.3%	91.3%	84.5%	87.9%
	KMA_σ^0	90.6%	79.1%	39.4%	80.7%	68.9%	69.7%

TABLE III

OVERALL ACCURACY AND KAPPA COEFFICIENT SUMMARY

Period	Method	Test N	Correct N	Kappa
Period 1	KMN_all	6257017	6006613	0.9195
	KMN_bt	6257017	6011344	0.9216
	KMN_σ^0	6257017	5546503	0.7573
	KMA_all	6257017	5769166	0.8539
	KMA_bt	6257017	5737639	0.8445
	KMA_σ^0	6257017	4914233	0.6074
Period 2	KMN_all	5921842	5692854	0.9207
	KMN_bt	5921842	5662060	0.9100
	KMN_σ^0	5921842	5225727	0.7615
	KMA_all	5921842	5659487	0.9097
	KMA_bt	5921842	5588649	0.8855
	KMA_σ^0	5921842	5113748	0.7278
Period 3	KMN_all	5270215	5004050	0.9130
	KMN_bt	5270215	5040446	0.9248
	KMN_σ^0	5270215	4542656	0.7525
	KMA_all	5270215	4931928	0.8892
	KMA_bt	5270215	4811758	0.8522
	KMA_σ^0	5270215	4210187	0.6622

in this range can obtain a higher accuracy rate and also avoid overuse of computational resources caused by using a too large K value.

For comparison, we chose two schemes of the two single-parameter methods at the same time and with K = 30. These specific results are shown in Tables II and III.

Table II shows that when K is set to 30, the accuracy rate of MYI in the two schemes (KMN_σ^0, KMA_σ^0) utilizing only σ^0 in time Periods 1 and 3 is relatively low, not exceeding 40%. Concurrently, its recall rate is less than 70%. The recognition rate of multiyear ice in the two scenarios using only brightness temperature (KMN_bt, KMA_bt) is significantly enhanced, with an accuracy rate of at least 80%. There is a noticeable correlation between the distribution results depicted in Figs. 5 and 8 and the characteristics observed in the present study. The distribution of σ^0 exhibits considerable overlap between FYI and MYI, making it challenging to distinguish between the two classes. Incorporation of the brightness-temperature data significantly enhances the capacity to distinguish between the ice types, with the accuracy for MYI exceeding 80% in all instances.

In Period 2, it can be seen that the joint use of brightness temperature and σ^0 has a role in the identification of sea ice. The accuracy rate of sea ice identification for both schemes using the BT and σ^0 data jointly is the highest among the six tested methods, with 95.5% and 98.4% accuracies, respectively. Also, the overall precision and recall were the highest. In the other two time periods, the combined use of the two methods during Periods 1 and 2 has much less impact on the overall results than during the sea ice melting Period 2.

To further quantify the uncertainty and reliability of the classification results, Table III summarizes the number of test samples, the number of correctly classified samples, and Cohen's kappa coefficient for the independent test subset at K = 30. Since the overall accuracy has already been reported in Tables II and III focuses on the sample statistics and kappa coefficient to provide an additional evaluation of the agreement between the classification results and the reference labels, especially under class imbalance conditions.

The results show that the BT-based and combined-feature methods generally achieved higher and more stable agreement than the σ^0-only methods. For the KMN method, KMN_bt obtained kappa coefficients of 0.9216, 0.9100, and 0.9248 in Periods 1, 2, and 3, respectively, which are comparable to or slightly higher than those of KMN_all. This indicates that the BT features provide strong discriminatory information for the considered sea ice classes. On the contrary, KMN_σ^0 showed lower kappa coefficients of 0.7573, 0.7615, and 0.7525 in the three periods.

The σ^0-only methods showed the largest uncertainty, especially for the KMA_σ^0 method. The kappa coefficients of KMA_σ^0 were 0.6074, 0.7278, and 0.6622 in Periods 1, 2, and 3, respectively, which were lower than those of the corresponding BT-based and combined-feature methods. This confirms that σ^0 alone is insufficient for robust FYI/MYI separation during the freezing periods. The classwise recall values shown in Table II further indicate that the performance degradation is mainly caused by classwise confusion rather than by the dominant class alone.

During Period 2, the classification was more stable because FYI and MYI were merged into a single sea ice class. This reduced the ambiguity associated with melt-season ice-type separation and led to relatively high classification accuracies in Table II and high kappa values in Table III for the BT-based and combined-feature methods. Overall, the additional kappa-based uncertainty assessment supports the conclusion that the combined use of active and passive observations improves classification robustness, while the FYI/MYI distinction remains the main source of uncertainty during the freezing periods.

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TABLE IV

AVERAGE SEA ICE-TYPE CLASSIFICATION ACCURACY FOR TWO STUDIES PRIOR TO THIS STUDY

	Zygmuntowska et al. 2013 (Bayesian)	Rinne et al. 2016 (KNN)	This study (KMN, multi-parameter)	This study (KMA, multi-parameter)
FYI	73.6%	90.0%	95.6%	92.3%
MYI	87.5%	64.4%	86.4%	90.0%

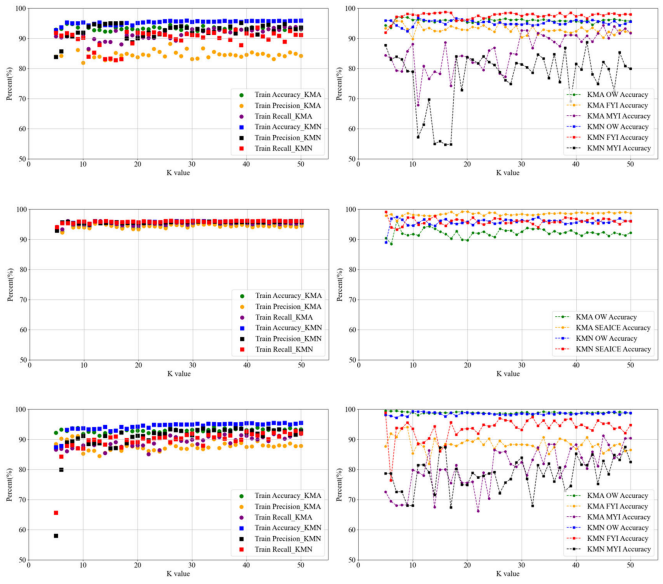


Fig. 11. Comparison of results of KMN and KMA methods.

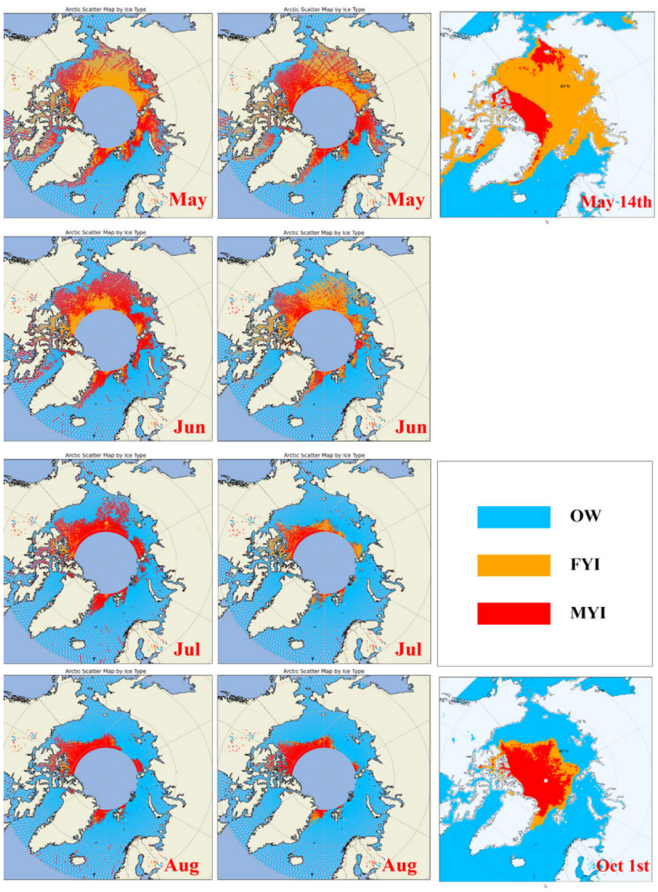


Fig. 12. Exploratory sea ice-type classification results for the summer months using the KMN method. The first and second columns show the classification results obtained by applying the three-class models trained with Period 1 and Period 3 data, respectively, to the melting-season data. The OSI-SAF sea ice-type products from May 14 and October 1 are used as qualitative References because corresponding summer sea ice-type products are unavailable.

V. DISCUSSION

Compared with the results of some prior studies (Table II), it is notable that the inclusion of passive radiometers has significantly enhanced the identification of ice types compared to these earlier studies.

In previous studies by Zygmuntowska et al. [13] and Rinne and Similä [31], only RA data were used for ice-type classification. They employed an RA with a higher ground resolution than HY-2B. However, our study's K-means results outperformed the results of these experiments (See Tables II and IV).

The overall analysis in Fig. 9 shows that the importance of the parameters depends strongly on the seasonal cycle and the specific classification objective. During the freezing season, the brightness-temperature parameters are important in all classifications, indicating their high reliability in identifying and distinguishing ice types at low temperatures. This is consistent with the known sensitivity of microwave brightness temperature to ice thickness and structure [32], [33], [34]. On the contrary, during the melting season, the importance of the parameters shifts toward those that capture surface scattering and roughness, such as Ku_σ^0 and C_σ^0, as the physical and radiative properties of the ice surface change with increasing temperature. The presence of water on the ice surface during melting alters the signal response, making the surface roughness-sensitive parameters more important for classification.

Although the quantitative classification in Period 2 was limited to the discrimination between OW and sea ice, we further conducted an additional experiment to examine whether sea ice-type information could still be inferred during the melting season. In this experiment, the three-class KMN models trained using the freezing-season data were applied to the summer months. In particular, the models trained with Period 1 and Period 3 data were used to classify the melting-season data from June to September. The results are shown in Fig. 12. It should be emphasized that the classification results in Fig. 12 were not trained using Period 2 data. Instead, they represent an exploratory transfer experiment from the freezing season to the melting season, aiming to evaluate the seasonal transferability and potential limitations of the proposed algorithm for summer sea ice-type classification.

During the melting season, the discrimination between FYI and MYI becomes much more difficult than in the freezing season. This is mainly because surface melt, wet snow, melt ponds, and changes in surface roughness can significantly modify the microwave brightness temperature and radar backscattering responses of sea ice. As a result, the radiometric and scattering contrasts between FYI and MYI are weakened, and the feature space learned from freezing-season

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samples may no longer be fully representative of the summer conditions. Therefore, in the main quantitative classification experiment for Period 2, only OW and sea ice were distinguished. The summer sea ice-type results shown in Fig. 12 should thus be interpreted as an exploratory classification attempt based on freezing-season models, rather than as a fully validated summer sea ice-type product.

As shown in Fig. 12, the classification results obtained using the Period 1 training set are generally more similar to the OSI-SAF sea ice-type distribution on May 14, while the results obtained using the Period 3 training set show better agreement with the OSI-SAF product on October 1, especially for September. This indicates that the summer classification results are strongly affected by the seasonal proximity between the training data and the target classification date. In other words, the model trained closer to the target season tends to produce a more reasonable spatial distribution. For example, the results for June are more consistent with the early season reference when the Period 1 model is used, whereas the September results are closer to the autumn reference when the Period 3 model is used. This phenomenon suggests that the microwave signatures of sea ice evolve continuously from the freezing season to the melting season and then back to the refreezing period, and that the applicability of a trained model depends strongly on the similarity between the training conditions and the target conditions.

However, large discrepancies can still be observed between the proposed classification results and the OSI-SAF sea ice-type product. These discrepancies are particularly evident in some marginal ice zones and coastal regions, where the proposed algorithm may identify FYI-like or MYI-like areas that are inconsistent with the OSI-SAF product. Several factors may contribute to these differences. First, the OSI-SAF products used here are not from the same dates as the summer classification results, because corresponding OSI-SAF sea ice-type products during the summer months are unavailable. The May 14 and October 1 products were therefore used only as qualitative references before and after the melting season. During the time interval between the reference dates and the classification dates, sea ice drift, deformation, melting, and redistribution may substantially change the spatial distribution of ice types. Second, the physical properties of sea ice during summer are highly variable, and melt ponds or wet snow can make MYI appear radiometrically similar to FYI, or even to OW in some cases. Third, the KMN model was trained using freezing-season feature distributions, and these distributions may shift considerably under summer melt conditions, leading to classification uncertainty when the model is transferred to the melting season.

Therefore, Fig. 12 should be regarded as a qualitative assessment of the potential and limitations of applying freezing-season sea ice-type models to summer conditions. The results suggest that the proposed algorithm can capture part of the seasonal evolution of sea ice-type distribution, especially when the training period is temporally close to the target month. Nevertheless, the reliable discrimination between FYI and MYI during the melting season remains challenging because of the strong seasonal modification of microwave

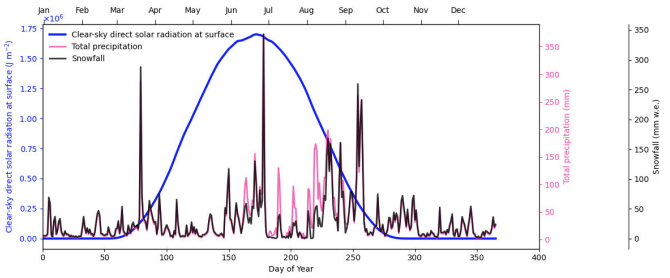


Fig. 13. Time series graph of snowfall, clear-sky direct solar radiation at the surface, and total precipitation.

signatures and the lack of coincident summer reference data. Further validation using independent summer sea ice-type observations, such as high-resolution SAR imagery or field-based ice-type information, would be necessary before robust summer FYI/MYI classification can be achieved.

In addition to the sensor-derived features, environmental conditions can also affect the microwave signatures of sea ice and therefore influence the classification results. In this study, ERA5 total precipitation, snowfall, and clear-sky direct solar radiation at the surface were used to support the interpretation of seasonal variations in classification performance [24]. These variables were not used to establish a direct quantitative prediction model of classification accuracy, but rather to explain possible physical sources of classification uncertainty.

As shown in Fig. 13, clear-sky direct solar radiation gradually increases from January to summer and reaches its maximum around day 174. From January to May, the increasing solar radiation can gradually promote surface warming and the early development of melt conditions. During June and July, stronger solar radiation and higher air temperature can further enhance surface melting, wet snow formation, and melt pond development. These processes modify the microwave emissivity of the sea ice surface and reduce the brightness-temperature contrast among different ice types [35], [36], [37]. This helps explain why FYI/MYI discrimination becomes more uncertain during the melting season and why only OW/sea ice classification was performed in Period 2.

Precipitation and snowfall also show clear seasonal variations. From January to May, precipitation and snowfall remain relatively smooth and at a lower level, suggesting that their influence on the physical properties of the sea ice surface is relatively limited during the stable freezing period. This is consistent with the better separability of OW, FYI, and MYI during Period 1. From August to September, sustained precipitation and snowfall may increase snow wetness and modify the snow layer over sea ice. Wet snow and rain-on-snow processes can change microwave emissivity, reduce or modify radar penetration, and alter radar backscatter and waveform responses [38], [39]. These effects may weaken the separability of both TB- and σ^0-based features and contribute to larger classification uncertainty during the late melting and early refreezing stages.

From October onward, snowfall increases markedly and reaches a high level in late October, while clear-sky direct solar radiation decreases rapidly and remains low until the following

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spring. Under these conditions, the influence of direct solar radiation on winter sea ice-type classification becomes limited, whereas changes in snow accumulation, snow wetness, and refrozen layers may still affect the passive microwave and RA responses. This may partly explain why the classification performance in Period 3 differs from that in Period 1, although both periods are generally considered freezing periods.

Overall, the seasonal behavior of these environmental variables is broadly consistent with the observed classification performance. During stable freezing months, the microwave signatures of OW, FYI, and MYI are more separable, and the classification results are generally more reliable. On the contrary, during melting and refreezing transition months, changes in surface wetness, snow properties, melt ponds, and radiation conditions reduce the separability of the input features and increase classification uncertainty [37], [38], [39], [40]. Therefore, the environmental variables discussed here provide a physical explanation for the seasonal differences in classification performance. However, a more rigorous quantitative analysis of the relationship between environmental variables and classification errors requires additional independent validation data and will be addressed in future work.

VI. Conclusion

This study classified Arctic ice types (AIT) using spaceborne active and passive microwave observations from the Chinese marine dynamic environment satellite HY-2B. The results demonstrate the capability of nadir-looking HY-2B observations for AIT classification over a full annual cycle in 2019. Although nadir observations have a narrower swath than scanning radiometers or scatterometers, they provide high along-track resolution of approximately 300–350 m owing to the high sampling frequency (~20 Hz) and satellite orbital speed (~7 km/s). Therefore, the HY-2B active–passive observation configuration shows potential for fine-scale along-track sea ice monitoring and targeted regional classification beneath the satellite ground tracks.

Two K-means-based methods, KMN and KMA, were developed using colocated active–passive nadir observations from HY-2B. KMN assigns samples according to the nearest labeled cluster center in Euclidean feature space, whereas KMA assigns samples to the labeled cluster with the maximum fit probability density. The results show that KMA generally performs better than KMN under more complex ice conditions, with an overall accuracy of about 90% for FYI/MYI classification. Sensitivity tests also indicate that low K values lead to unstable classification accuracy, while K values between 25 and 35 produce more stable results. Environmental analysis further suggests that precipitation and snowfall have limited influence during stable winter conditions, but can affect surface emissivity and radar backscatter during early growth and transition periods.

The combination of passive microwave brightness temperatures and active RA backscatter improves sea ice-type classification compared with the use of individual sensor features alone. In particular, the integrated active–passive approach improves the classification accuracy by more than 5% compared with classifications based only on σ^0, demonstrating the added value of complementary HY-2B observations. The results also indicate potential for monitoring summer ice evolution; however, reliable summer sea ice-type validation remains limited. Future work should therefore focus on developing appropriate summer validation datasets and further evaluating the classification performance during the melt season.

After the cluster-label relationship and pdf parameters are determined from the training dataset, the KMA method can be applied using only collocated HY-2B active–passive observations as input features. Its relatively simple structure and low computational cost make it potentially suitable for efficient along-track sea ice-type classification. With continued HY-2 observations and further multiyear validation, this approach may support future operational applications in polar sea ice monitoring, particularly for targeted regions beneath satellite ground tracks.

Future work will focus on improving the robustness and temporal transferability of the classification method across different seasonal conditions. As this study was limited to the complete 2019 HY-2B active–passive dataset, future work will extend the method to additional years when complete calibrated and quality-controlled datasets become available. More detailed local weather information and multiseason observations may also be incorporated to better account for regional differences in melt and freeze timing across the Arctic. One possible direction is to combine the complementary strengths of KMN and KMA in a hybrid or adaptive framework to reduce ambiguous classifications. In addition, the fusion of nearly simultaneous Earth observation data, such as SAR imagery, may further improve the spatial resolution and accuracy of sea ice type and ice/water classification. More detailed ice-age estimation beyond the binary FYI/MYI classification will also be investigated in future studies.

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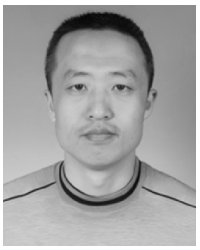
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